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Next item →

1. Suppose that we wish to track the 1-dimensional state of a system modeled by:

1 / 1 point

$$x_{k+1} = \alpha x_k^2 + w_k$$
$$z_k = \sin(x_k) + v_k,$$

where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$. What is $f(x_k \mid x_{k-1})$?

- ☐ $\mathcal{N}(0, \sigma_w^2)$.
- ☐ $\mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$.
- ☒ $\mathcal{N}(\alpha x_{k-1}^2, \sigma_w^2)$.
- ☐ $\mathcal{N}(0, \sigma_v^2)$.
- ☐ $\mathcal{N}(0.1 x_k^3, \sigma_v^2)$.
- ☐ $\mathcal{N}(\sin(x_k), \sigma_v^2)$.

✔ Correct
Yes, this is correct.

2. Suppose that we wish to track the 1-dimensional state of a system modeled by:

1 / 1 point

$$x_{k+1} = \alpha x_k^2 + w_k$$
$$z_k = \sin(x_k) + v_k,$$

where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$. What is $f(z_k \mid x_k)$?

- ☐ $\mathcal{N}(0, \sigma_w^2)$.
- ☐ $\mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$.
- ☐ $\mathcal{N}(\alpha x_{k-1}^2, \sigma_w^2)$.
- ☐ $\mathcal{N}(0, \sigma_v^2)$.
- ☐ $\mathcal{N}(0.1 x_k^3, \sigma_v^2)$.
- ☒ $\mathcal{N}(\sin(x_k), \sigma_v^2)$.

✔ Correct
Yes, this is correct.

3. How do we compute $f(z_k \mid \mathbb{Z}_{k-1})$ in the denominator of the time-update equation for $f(x_k \mid \mathbb{Z}_k)$?

1 / 1 point

- ☐ We apply the law of total probability and write this as an integral that we can approximate using either the rectangular rule or the trapezoid rule.
- ☒ We don't. It is needed only as a normalization factor that we can determine in other ways.

✔ Correct
Yes, this is the approach that we will take.